SAP ERP Implementation

Project Blueprint & Documentation

NovaTex Industries Pvt. Ltd.

A Fictitious Textile & Apparel Manufacturing Enterprise

Submitted By

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1. Problem Statement

NovaTex Industries Pvt. Ltd. is a mid-sized textile and apparel manufacturer based in Surat, Gujarat, India. The company operates three manufacturing plants across western India and distributes its products through a nationwide dealer network covering 18 states. Despite steady revenue growth of 22% year-on-year, the company continues to face severe operational challenges stemming from fragmented, legacy information systems.

The core problems identified through a pre-implementation study are:

- Financial data is maintained in three separate standalone accounting packages, making consolidated reporting impossible in real time.
- Procurement and inventory management rely on disconnected spreadsheets, causing frequent stock-outs and excess raw material accumulation.
- Sales order processing is handled manually via phone and email, leading to order entry errors, delayed invoicing, and poor customer satisfaction.
- There is no integration between the Finance, Purchase, and Sales departments, resulting in payment delays, duplicate vendor payments, and revenue leakage.
- Audit trails are non-existent, creating compliance risks under GST and corporate tax regulations.

2. Proposed Solution

The proposed solution is the implementation of SAP S/4HANA (on-premise), configured specifically for the textile manufacturing industry. The implementation spans three core SAP functional modules — Financial Accounting (FI), Materials Management (MM), and Sales and Distribution (SD) — chosen to address each identified problem area with a unified, real-time data backbone.

Module	Key Features Implemented
SAP FI	Company code setup, Chart of Accounts, GL Accounting, Accounts Payable, Accounts Receivable, Tax configuration (GST)
SAP MM	Material Master, Vendor Master, Purchase Requisition, Purchase Order, Goods Receipt, Invoice Verification (Procure-to-Pay cycle)
SAP SD	Customer Master, Sales Area, Sales Order, Delivery, Billing (Order-to-Cash cycle)
Integration	FI-MM (automatic account determination on GR/IR), FI-SD (billing to AR posting), MM-SD (stock availability check)

3. Tech Stack

ERP Platform	SAP S/4HANA (on-premise, 2023 release)
Database	SAP HANA in-memory database

Functional Modules	FI (Financial Accounting), MM (Materials Management), SD (Sales & Distribution)
Configuration Tool	SAP Implementation Guide (IMG / SPRO transaction)
Reporting	SAP Fiori Analytical Apps, SAP Report Painter
Integration	ALE/IDOC for inter-system data exchange; Standard SAP module integration (FI-MM, FI-SD)
Development	ABAP (custom reports and user exits where standard config is insufficient)

4. Company Blueprint — NovaTex Industries Pvt. Ltd.

NovaTex Industries Pvt. Ltd. is a fictitious mid-sized company engaged in the manufacturing and distribution of cotton, polyester-blend, and technical textiles. Founded in 2005 and headquartered in Surat, Gujarat, the company employs approximately 1,800 people and reports an annual turnover of INR 480 crore.

Industry	Textile & Apparel Manufacturing (NIC Code: 1311)
Headquarters	Surat, Gujarat, India
Plants	Surat (Main), Ahmedabad (Knitting), Pune (Finishing & Export)
Product Lines	Cotton Fabrics, Polyester Blends, Technical Textiles, Readymade Garments
Distribution	Pan-India (18 states); Export to UAE and Bangladesh
Customer Base	B2B — Wholesale Distributors, Retail Chains, Export Houses
Financial Year	April 1 to March 31 (Indian fiscal year)
Currency	INR (Indian Rupee)

Organizational Structure in SAP:

SAP Object	ID	Description
Client	100	NovaTex Group (SAP Production Client)
Company Code	NTXL	NovaTex Industries Pvt. Ltd.
Plant	NTX1	Surat Manufacturing Plant (Main)
Plant	NTX2	Ahmedabad Knitting Unit
Plant	NTX3	Pune Finishing & Export Unit
Storage Location	SL01	Raw Material Store — Surat
Storage Location	SL02	Finished Goods Store — Surat
Storage Location	SL03	Spare Parts Store — Surat
Purchasing Org.	PO01	Central Purchasing — NovaTex
Purchasing Group	PG01	Raw Material Procurement
Purchasing Group	PG02	Packaging & Consumables
Sales Organization	SO01	NovaTex Sales India
Distribution Channel	DC01	Wholesale
Distribution Channel	DC02	Direct Export
Division	DV01	Cotton & Blended Fabrics

Division	DV02	Technical Textiles
Sales Office	SOF1	Mumbai Sales Office
Sales Office	SOF2	Delhi Sales Office
Chart of Accounts	NTCA	NovaTex Chart of Accounts
Fiscal Year Variant	V3	April–March (Indian fiscal year)
Posting Period Variant	NTPP	NovaTex Posting Periods

Organizational Interconnections:

- Company Code NTXL is assigned to Chart of Accounts NTCA and Fiscal Year Variant V3, forming the core financial backbone.
- Plants NTX1, NTX2, and NTX3 are all assigned to Company Code NTXL, ensuring all manufacturing activities post to the same legal entity.
- Purchasing Organization PO01 is assigned to Company Code NTXL and also linked to all three plants (plant-specific purchasing is allowed).
- Sales Organization SO01 is assigned to Company Code NTXL, Distribution Channels DC01 and DC02, and Divisions DV01 and DV02.
- The Sales Area (combination: SO01 + DC01 + DV01) represents the standard domestic wholesale sales channel for fabrics.
- Plant NTX1 is the delivering plant for domestic Sales Organization SO01, connecting SD with MM for goods issue and stock management.

5. SAP FI — Financial Accounting Configuration

5.1 Company Code Creation (Transaction: OX02)

The Company Code is the smallest organizational unit for which a complete, self-contained set of accounts can be drawn up. For NovaTex, Company Code NTXL is created with the following IMG path and settings:

IMG Path: SPRO > Enterprise Structure > Definition > Financial Accounting > Edit, Copy, Delete, Check Company Code

Field	Value
Company Code	NTXL
Company Name	NovaTex Industries Pvt. Ltd.
City	Surat
Country	IN (India)
Currency	INR
Language	EN
Address (Street)	Plot No. 45, Sachin GIDC, Surat, Gujarat – 394230

5.2 Chart of Accounts (Transaction: OB13)

The Chart of Accounts (COA) defines the structure and list of GL accounts used. NovaTex uses a custom COA 'NTCA' modelled on the Indian standard accounting structure.

IMG Path: SPRO > Financial Accounting > General Ledger Accounting > GL Accounts > Master Data > Preparations > Edit Chart of Accounts List

Account Range	Account Group	Description
100000–199999	ASSET	Fixed Assets & Current Assets
200000–299999	LIAB	Liabilities & Provisions
300000–399999	EQUITY	Share Capital & Reserves
400000–499999	REVN	Revenue & Sales Accounts
500000–599999	COGS	Cost of Goods Sold / Direct Costs
600000–699999	EXPN	Operating Expenses

700000–799999	TAX	Tax Accounts (GST Input/Output)
800000–899999	RECON	Reconciliation Accounts (AP/AR)

5.3 Fiscal Year Variant & Posting Period Variant

Configuration Item	ID / Value	Details
Fiscal Year Variant	V3	April 1 – March 31 (12 periods + 4 special periods)
Period 1	April	Month 1 of financial year
Period 12	March	Month 12 of financial year
Special Periods	13–16	Year-end audit and closing adjustments
Posting Period Variant	NTPP	Assigned to Company Code NTXL; controls open/close periods

IMG Path (Fiscal Year): SPRO > Financial Accounting > Financial Accounting Global Settings > Fiscal Year > Maintain Fiscal Year Variant

IMG Path (Posting Period): SPRO > Financial Accounting > Financial Accounting Global Settings > Document > Posting Periods > Define Variants for Open Posting Periods

5.4 GL Account Creation (Transaction: FS00)

Key GL accounts created in Company Code NTXL are listed below. Each account is maintained at two levels: Chart of Accounts level (general definition) and Company Code level (control parameters specific to NTXL).

GL Account No.	Description	Account Type	Recon. Acct
100100	HDFC Bank Current A/C	Balance Sheet	No
130000	Trade Receivables (AR)	Recon. Account	Customers
200000	Trade Payables (AP)	Recon. Account	Vendors
310000	Share Capital	Balance Sheet	No
400100	Domestic Sales — Fabric	P&L;	No
500100	Raw Material Consumption	P&L;	No
700100	GST Input Tax (18%)	Balance Sheet	No
700200	GST Output Tax (18%)	Balance Sheet	No

5.5 Customer & Vendor Master Setup

Customer and Vendor Masters in SAP FI have three views: General Data (client-level), Company Code Data (financial parameters), and Sales/Purchasing Data (module-specific). For NovaTex, the reconciliation accounts 130000 (Customers) and 200000 (Vendors) are assigned at the company code level.

Parameter	Customer (FI View)	Vendor (FI View)
Account Group	CUST (Sold-to Party)	VEND (External Vendor)
Recon. GL Account	130000	200000
Payment Terms	NT30 (Net 30 days)	NT45 (Net 45 days)
Dunning Procedure	DNOV (Monthly dunning)	N/A
Withholding Tax	TDS Section 194C	TDS Section 194C
Transaction (Create)	FD01	FK01

6. SAP MM — Materials Management Configuration

6.1 Material Master Creation (Transaction: MM01)

Material Master is the central repository of all material-related data in SAP. For NovaTex, materials are categorized into Raw Materials, Semi-Finished Goods, Finished Goods, and Packaging. Each material type has its own number range and organizational views.

IMG Path: SPRO > Logistics — General > Material Master > Basic Settings > Material Types > Define Attributes of Material Types

Material Code	Description	Mat. Type	UoM	Plant	Valuation Class
NTX-RM-001	Raw Cotton (Grade A)	ROH	KG	NTX1	3000 (Raw Mat.)
NTX-RM-002	Polyester Fibre	ROH	KG	NTX1	3000
NTX-RM-003	Dyes & Chemicals	ROH	LTR	NTX1	3001
NTX-SFG-001	Greige Fabric (100cm)	HALB	MTR	NTX1	7900 (Semi-FG)
NTX-FG-001	Cotton Shirting 2/40s	FERT	MTR	NTX1	7920 (Fin. Goods)
NTX-FG-002	Poly-Cotton Suiting	FERT	MTR	NTX1	7920
NTX-PKG-001	Cardboard Carton (Large)	VERP	EA	NTX1	3100 (Packaging)

Key views maintained in Material Master: Basic Data 1, Basic Data 2, Purchasing, MRP 1 & 2, Accounting 1 (valuation), Storage, and Sales: Sales Org. Data.

6.2 Vendor Master (Transaction: MK01 / XK01)

Vendor Master data is shared between MM (purchasing view) and FI (accounting view). Central creation via XK01 ensures data consistency across both modules.

Vendor Code	Name	City	Account Group	Purchasing Org.
V001	Gujarat Cotton Traders	Surat	VEND	PO01
V002	Reliance Industries Ltd.	Mumbai	VEND	PO01
V003	BASF India Pvt. Ltd.	Pune	VEND	PO01
V004	Arvind Packaging Co.	Baroda	VEND	PO01

6.3 Purchasing Organization Setup

IMG Path: SPRO > Enterprise Structure > Definition > Materials Management > Maintain Purchasing Organization

Step	IMG Path / Transaction	Action
1	OX08 — Define Purch. Org.	Create PO01 'Central Purchasing — NovaTex'
2	OX01 — Assign Purch. Org. to Company Code	Assign PO01 → NTXL
3	OX17 — Assign Purch. Org. to Plant	Assign PO01 → NTX1, NTX2, NTX3
4	OME4 — Define Purchasing Groups	Create PG01 (Raw Mat.), PG02 (Packaging)
5	OMFI — Number Ranges for PO	Set PO number range 4500000000–4599999999

6.4 Procure-to-Pay (P2P) Process Flow

The P2P cycle covers the complete procurement lifecycle from identifying a material requirement to settling the vendor invoice. Each step below includes the SAP transaction code and the accounting/inventory impact.

Step	Activity	T-Code	Impact
1	Purchase Requisition created by Production Planning team for Raw Cotton (500 KG)	ME51N	No financial posting; inventory not affected
2	Purchasing creates Purchase Order for Vendor V001 @ INR 68/KG (Total: INR 34,000)	ME21N	No financial posting; PO committed value updated
3	Goods Receipt posted at Plant NTX1, Storage Location SL01	MIGO	Dr. Raw Material Stock INR 34,000 | Cr. GR/IR Clearing INR 34,000
4	Logistics Invoice Verification — Vendor invoice matched to PO and GR	MIRO	Dr. GR/IR Clearing INR 34,000 | Cr. Vendor Payable INR 34,000 (+ GST)
5	Automatic Payment Run to Vendor via F110	F110	Dr. Vendor Payable | Cr. Bank Account

7. SAP SD — Sales & Distribution Configuration

7.1 Sales Area Setup

A Sales Area is the unique combination of Sales Organization, Distribution Channel, and Division. It is the key structural element controlling sales transactions in SAP SD.

IMG Path: SPRO > Enterprise Structure > Assignment > Sales and Distribution > Set up Sales Area

Sales Org.	Dist. Channel	Division	Sales Area Description
SO01	DC01	DV01	Domestic Wholesale — Cotton & Blended Fabrics
SO01	DC01	DV02	Domestic Wholesale — Technical Textiles
SO01	DC02	DV01	Export — Cotton & Blended Fabrics
SO01	DC02	DV02	Export — Technical Textiles

7.2 Customer Master (Transaction: XD01)

Customer Master in SD contains general data (name, address), company code data (payment terms, reconciliation account), and sales area data (sales district, pricing procedure, incoterms). A sample customer master for a key domestic buyer is:

Field	Value
Customer Code	C001
Customer Name	Bombay Textile Mart Pvt. Ltd.
City / Region	Mumbai, Maharashtra
Account Group	CUST (Sold-To Party)
Reconciliation Account	130000 (Trade Receivables)
Payment Terms	NT30 — Net 30 days
Sales Area	SO01 / DC01 / DV01
Pricing Procedure	RVAA01 (Standard SD Pricing)
Incoterms	EXW — Ex-Works Surat
Delivering Plant	NTX1 — Surat Main Plant

7.3 Order-to-Cash (O2C) Process Flow

Step	Activity	T-Code	Impact
1	Customer C001 places order for 2,000 MTR of NTX-FG-001 @ INR 95/MTR	VA01	Sales Order created; no financial posting; availability check run
2	Delivery document created; picking and packing done at NTX1, SL02	VL01N	No financial posting; transfer order created in WM (if applicable)
3	Post Goods Issue (PGI) — goods dispatched to customer	VL02N	Dr. COGS (500100) INR 1,50,000 | Cr. Inventory FG INR 1,50,000
4	Billing document (Invoice) created and sent to customer	VF01	Dr. Trade Receivable (130000) INR 2,24,440 | Cr. Revenue (400100) INR 1,90,000 | Cr. GST Output Tax 18% (700200) INR 34,200 + cess
5	Customer payment received and cleared against open invoice	F-28	Dr. Bank Account | Cr. Trade Receivable (130000)

8. End-to-End Integration: FI–MM–SD

One of the most significant advantages of SAP ERP is real-time, seamless integration across functional modules. The following diagram and explanation describe how FI, MM, and SD interact in NovaTex's business processes.

FI–MM Integration (Procure-to-Pay):

- When a Goods Receipt (MIGO) is posted in MM, SAP automatically creates an accounting document in FI: debit to Raw Material Stock and credit to GR/IR Clearing Account — no manual journal required.
- During Invoice Verification (MIRO), the GR/IR clearing is reversed, and the Vendor Payable (AP) account is credited. This is controlled by the Automatic Account Determination configuration (T-Code OBYC).
- Payment processing (F110) debits the Vendor Payable and credits the Bank GL account, completing the liability cycle.
- Valuation of inventory in FI is driven by the Valuation Class assigned in Material Master, mapped to GL accounts via OBYC — ensuring consistent stock valuation.

FI–SD Integration (Order-to-Cash):

- Post Goods Issue (VL02N) in SD triggers an automatic accounting entry in FI: COGS is debited and Inventory is credited, reducing stock in real time.
- Billing (VF01) creates a Customer (AR) document in FI: Trade Receivable is debited; Revenue and GST Output Tax accounts are credited. The Revenue Account Determination (T-Code VKOA) controls this mapping.
- Incoming customer payments (F-28) clear the open AR items, completing the revenue recognition cycle.
- Credit management in SD checks the customer's credit limit (configured in FI-AR) before allowing Sales Order confirmation.

MM–SD Integration:

- During Sales Order creation (VA01), the Availability Check (configured in MM) queries real-time stock at plant NTX1 to confirm whether the requested quantity can be delivered by the required date.
- Transfer of Requirements ensures that confirmed Sales Orders automatically generate requirements in MRP (Materials Requirements Planning), triggering procurement if stock is insufficient.

9. Unique Points of the Project

1. Indian GST-Compliant Tax Configuration

The FI configuration includes CGST, SGST, and IGST tax codes with correct GL account mapping, reflecting real-world Indian tax compliance requirements rather than a generic international setup.

2. Multi-Plant Architecture with Centralized Purchasing

Three geographically distributed plants (Surat, Ahmedabad, Pune) are managed under a single Purchasing Organization, demonstrating a realistic corporate procurement structure.

3. Dual Distribution Channel (Domestic + Export)

The Sales Area setup distinguishes between domestic wholesale (DC01) and export channels (DC02), each with separate pricing procedures, incoterms, and billing types — reflecting actual business complexity.

4. Valuation Class-Driven Automatic Account Determination

Materials are assigned Valuation Classes that map directly to specific GL accounts, eliminating manual postings on every inventory movement and ensuring real-time financial accuracy.

5. End-to-End Procure-to-Pay and Order-to-Cash with Accounting Entries

Unlike many academic projects that stop at the functional transaction level, this project explicitly traces the accounting impact of every business event across both cycles.

10. Future Improvements

1. SAP PP (Production Planning) Integration

Extending the project to include SAP PP would connect production orders with raw material consumption (MM) and finished goods production (SD), completing the full manufacturing cycle.

2. SAP QM (Quality Management)

Adding QM would allow quality inspection lots at Goods Receipt and Goods Issue, ensuring only quality-passed materials enter production and only approved finished goods are dispatched to customers.

3. SAP BW/4HANA Analytics

Integrating SAP BW/4HANA would provide real-time dashboards on procurement costs, inventory turnover, sales revenue by region, and working capital metrics — enabling data-driven decision making.

4. Electronic Data Interchange (EDI) with Key Customers

Automating Sales Order creation via EDI would eliminate manual order entry, reduce errors, and accelerate the Order-to-Cash cycle for key wholesale customers.

5. SAP Fiori Mobile Apps

Deploying SAP Fiori apps would enable plant managers, purchase officers, and sales executives to approve requisitions, check stock, and monitor sales orders on mobile devices, improving operational agility.

6. GST Return Filing Automation

Integrating SAP with the GSTN (GST Network) via APIs would automate GSTR-1, GSTR-3B, and input tax credit reconciliation, reducing compliance effort.

This document has been prepared as an academic project submission for the B. Tech CSE (SAP Data Analytics) programme. All company names, vendor details, customer data, and financial figures used herein are entirely fictitious and are used solely for academic demonstration purposes.

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